

Part 1:

The screenshot shows the SQL Studio interface. On the left, the 'SCHEMAS' pane lists databases: helpdesk, my_engineeringdatabase, premiere, sakila, sys, test_db, and world. The 'helpdesk' database is selected, showing its tables, views, stored procedures, and functions. The 'Connection Details' pane shows the connection to 'Local instance' on 'localhost' port '3306' using 'root' user. The 'Server' is 'MySQL Commu' version '8.4.10', and the 'Connector' is 'C++ 9.6.0'.

The main editor shows a script named 'Helpdesk insert_v03' with the following SQL commands:

```
1 # Helpdesk insert_v03
2 use helpdesk;
3 select * from employee;
4
5 INSERT INTO `helpdesk`.`employee` (`FirstName`, `LastName`, `Location`) VALUES
6 ('Will', 'Smith', 'MD');
```

The 'Result Grid' shows the output of the script. It contains two tables: 'employee' and 'tickets'.

idTickets	Description	Duration	Priority	Status	Employee_idEmployee
1	Malware	33	3-Low	Completed	1
2	Virus	17	1-High	In Progress	7
3	Hardware	NULL	3-Low	Not Started	NULL
4	Malware	28	1-High	Completed	7
5	Virus	14	2-Medium	Completed	5
6	Virus	20	2-Medium	Completed	2

The 'Output' pane shows the execution log:

#	Time	Action	Message
82	19:30:26	use helpdesk	0 row(s) affected
83	19:30:26	select * from employee LIMIT 0, 1000	0 row(s) returned
84	19:30:26	INSERT INTO `helpdesk`.`employee` (`FirstName`, `LastName`, `Location`) VALUES ('Will', 'S...	7 row(s) affected Records: 7 Duplicates
85	19:30:26	select * from employee LIMIT 0, 1000	7 row(s) returned
86	19:30:26	select * from tickets LIMIT 0, 1000	0 row(s) returned
87	19:30:26	INSERT INTO `helpdesk`.`tickets` (`Description`, `Duration`, `Priority`, `Status`, `Employee_idE...	30 row(s) affected Records: 30 Duplicate
88	19:30:26	select * from tickets LIMIT 0, 1000	30 row(s) returned

The screenshot shows the SQL Studio interface. The 'SCHEMAS' pane is the same as in the previous screenshot. The main editor shows a query:

```
1 select count(*) from helpdesk.employee;
2
3
```

The 'Result Grid' shows the output of the query:

count(*)
7

The screenshot shows the SQL Studio interface. The 'SCHEMAS' pane is the same as in the previous screenshot. The main editor shows a query:

```
2 select count(*) from helpdesk.tickets;
3
```

The 'Result Grid' shows the output of the query:

count(*)
30

Part 2-1

SCHEMAS

Filter objects

Columns

- ◆ idEmployee
- ◆ FirstName
- ◆ LastName
- ◆ Location

Indexes

Foreign Keys

Triggers

tickets

Columns

- ◆ idTickets
- ◆ Description
- ◆ Duration
- ◆ Priority

Administration

Schemas

Information

Limit to 1000 rows

```

1 select e.FirstName, e.LastName, t.idTickets, t.description, t.duration
2 from employee as e
3 Inner join tickets as t
4 on e.idEmployee=t.Employee_idEmployee
5 order by e.FirstName,
6 t.Duration;
7

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content: ☒

	FirstName	LastName	idTickets	description	duration
▶	Ben	Johnson	11	Malware	7
	Ben	Johnson	12	Malware	8
	Ben	Johnson	16	Hardware	12
	Ben	Johnson	19	Malware	15
	Ben	Johnson	21	Malware	16
	Ben	Johnson	6	Virus	20
	Ben	Johnson	17	Malware	33
	Ben	Johnson	28	Malware	39

Result 2
×

Output

Action Output
▼

#	Time	Action	Message
✓ 105	19:35:11	select count(*) from helpdesk.tickets LIMIT 0, 1000	1 row(s) returned
✗ 106	21:08:11	select e.FirstName, e.LastNanne, t.idTickets, t.description, t.duration from employee as e Inner...	Error Code: 1054. Unknc
✓ 107	21:09:19	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner j...	25 row(s) returned
✓ 108	21:11:14	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner j...	25 row(s) returned

Connection Details

Name: Local instance
Host: localhost
Port: 3306
Login User: root
Current User: root@localhost
SSL cipher: SSL not used

Server

Product: MySQL Commu
Version: 8.4.10

Connector

Version: C++ 9.6.0

Object Info

Session

2-2

The screenshot displays the MySQL Workbench environment. On the left, the 'Navigator' pane shows the 'CHEMAS' database with a tree view of its objects: Columns (idEmployee, FirstName, LastName, Location), Indexes, Foreign Keys, Triggers, and a 'tickets' table with its own Columns (idTickets, Description, Duration, Priority). Below this, the 'Administration' and 'Schemas' tabs are visible, along with 'Connection Details' (Name: Local instance, Host: localhost, Port: 3306, Login User: root, Current User: root@localhost, SSL cipher: SSL not used) and 'Server' information (Product: MySQL Commu, Version: 8.4.10, Connector Version: C++ 9.6.0).

The main workspace is divided into three panes. The top pane shows the SQL editor with a query:


```

1 select e.FirstName, e.LastName, t.idTickets, t.description, t.duration
2 from employee as e
3 left join tickets as t
4 on e.idEmployee=t.Employee_idEmployee
5 order by e.FirstName,
6 t.Duration;
    
```

 The middle pane displays the 'Result Grid' with a table of results:

	FirstName	LastName	idTickets	description	duration
▶	Ben	Johnson	11	Malware	7
	Ben	Johnson	12	Malware	8
	Ben	Johnson	16	Hardware	12
	Ben	Johnson	19	Malware	15
	Ben	Johnson	21	Malware	16
	Ben	Johnson	6	Virus	20
	Ben	Johnson	17	Malware	33
	Ben	Johnson	28	Malware	39

 The bottom pane shows the 'Output' tab with a log of actions:

#	Time	Action	Message
✗ 106	21:08:11	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner...	Error Code: 1054. Unl
✓ 107	21:09:19	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner j...	25 row(s) returned
✓ 108	21:11:14	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner j...	25 row(s) returned
✓ 109	21:17:01	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e left joi...	27 row(s) returned

2-3

HEMAS

Filter objects

- LastName
- Location
- Indexes
- Foreign Keys
- Triggers
- tickets
 - Columns
 - idTickets
 - Description
 - Duration
 - Priority
 - Status
 - Employee_idEmpl
 - Indexes

Administration Schemas

Information

Connection Details

Name: Local instance
Host: localhost
Port: 3306
Login User: root
Current User: root@localhost
SSL cipher: SSL not used

Server

Product: MySQL Commu
Version: 8.4.10

Connector

Version: C++ 9.6.0

Object Info Session

```

1 • select t.idTickets,t.Description,t.priority,t.status,e.FirstName
2   from employee as e
3   right join tickets as t
4   on e.idEmployee=t.Employee_idEmployee
5   order by t.Status,
6   t.idTickets;

```

Result Grid

	idTickets	Description	priority	status	FirstName
1	Malware	3-Low	Completed	Will	
4	Malware	1-High	Completed	Giriraj	
5	Virus	2-Medium	Completed	Elena	
6	Virus	2-Medium	Completed	Ben	
9	Malware	3-Low	Completed	Will	
10	Hardware	2-Medium	Completed	Giriraj	
11	Malware	3-Low	Completed	Ben	
13	Virus	3-Low	Completed	Giriraj	

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 108	21:11:14	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e Inner j...	25 row(s) returned
✓ 109	21:17:01	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e left joi...	27 row(s) returned
✓ 110	21:28:51	select t.idTickets,t.Description,t.priority,t.status from employee as e right join tickets as t on e.id...	30 row(s) returned
✓ 111	21:33:12	select t.idTickets,t.Description,t.priority,t.status,e.FirstName from employee as e right join ticket...	30 row(s) returned

2-4

SCHEMAS

Filter objects

helpdesk

- Tables
 - employee
 - Columns
 - idEmployee
 - FirstName
 - LastName
 - Location
 - Indexes
 - Foreign Keys
 - Triggers
 - tickets
 - Columns
 - idTickets

Administration Schemas

Information

Connection Details

Name: Local instance
Host: localhost
Port: 3306
Login User: root
Current User: root@localhost
SSL cipher: SSL not used

Server

Product: MySQL Commu
Version: 8.4.10

Connector

Version: C++ 9.6.0

Object Info Session

```

1 • select t.idTickets,t.Description,t.duration,t.priority,t.status,e.FirstName
2   from employee as e
3   inner join tickets as t
4   on e.idEmployee=t.Employee_idEmployee
5   where status='in progress' and duration >= 20;

```

Result Grid

	idTickets	Description	duration	priority	status	FirstName
25	Internet	36	2-Medium	In Progress	Diamond	
8	Malware	27	2-Medium	In Progress	Giriraj	

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 109	21:17:01	select e.FirstName, e.LastName, t.idTickets, t.description, t.duration from employee as e left joi...	27 row(s) returned
✓ 110	21:28:51	select t.idTickets,t.Description,t.priority,t.status from employee as e right join tickets as t on e.id...	30 row(s) returned
✓ 111	21:33:12	select t.idTickets,t.Description,t.priority,t.status,e.FirstName from employee as e right join ticket...	30 row(s) returned
✓ 112	22:00:22	select t.idTickets,t.Description,t.duration,t.priority,t.status,e.FirstName from employee as e inner...	2 row(s) returned

Navigator: M3 - helpdesk-setup M3 - helpdesk-insert SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8*

SCHEMAS

Filter objects

- LastName
- Location
- Indexes
- Foreign Keys
- Triggers
- tickets
 - Columns
 - idTickets
 - Description
 - Duration
 - Priority
 - Status
 - Employee_idEmpl
 - Indexes

Administration Schemas

Information

Connection Details

Name: Local instance
 Host: localhost
 Port: 3306
 Login User: root
 Current User: root@localhost
 SSL cipher: SSL not used

Server

Product: MySQL Commu
 Version: 8.4.10

Connector

Version: C++ 9.6.0

Object Info Session

```

1 • select t.Description,t.duration,t.idTickets,t.status,e.FirstName,e.location,t.priority
2   from employee as e
3  right join tickets as t
4   on e.idEmployee=t.Employee_idEmployee
5  where t.Priority='1-High' and (t.Description='Malware' or e.Location='MD')
6   order by t.Description asc,
7         t.Duration desc,
8         t.idTickets asc;
9

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Description	duration	idTickets	status	FirstName	location	priority
▶	Hardware	20	30	Completed	Will	MD	1-High
	Malware	28	4	Completed	Giriraj	CA	1-High
	Malware	15	19	Completed	Ben	MD	1-High
	Malware	8	12	In Progress	Ben	MD	1-High
	Malware	NULL	24	Not Started	NULL	NULL	1-High
	Malware	NULL	29	Not Started	NULL	NULL	1-High

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 125	22:22:26	select t.Description,t.duration,t.idTickets,t.status,t.priority FROM tickets as t LIMIT 0, 1000	30 row(s) returned
✓ 126	22:22:56	select t.Description,t.duration,t.idTickets,t.status,e.FirstName,e.location,t.priority from employee...	5 row(s) returned
✓ 127	22:25:40	select t.Description,t.duration,t.idTickets,t.status,e.FirstName,e.location,t.priority from employee...	6 row(s) returned